

Metallurgy

Amalgam : An alloy in which one of the component metals is mercury, is called amalgam.

In alloys, the chemical properties of the component elements are retained but certain physical properties are improved.

Compounds of metal and non-metal and their uses :

- 1. Ferrous oxide (FeO)** : In green glass, Ferrous salt.
- 2. Ferric oxide (Fe_3O_4)** : In electroplating of ornaments and formation of ferric slat
- 3. Ferrous sulphate ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$)** : In dye industry, and Mohr's salt
- 4. Ferric hydroxide [$\text{Fe}(\text{OH})_3$]** : In laboratory reagent and in making medicines.
- 5. Iodine (I_2)** : (i) As antiseptic, (ii) In making tincture of iodine.
- 6. Bromine (Br_2)** : (i) In dye industry (ii) As laboratory reagent
- 7. Chlorine (Cl_2)** : In the formation of (i) Mustard gas (ii) Bleaching powder
- 8. Hydrochloric acid (HCl)** : In the formation of aquaregia ($3 \text{HCl} : 1 \text{HNO}_3$) and dyes
- 9. Sulphuric acid (H_2SO_4)** : (i) As a reagent (ii) In purification of petroleum (iii) In lead storage battery.
- 10. Sulphur dioxide (SO_2)** : (i) As oxidants & reductants (ii) As bleaching agent
- 11. Hydrogen Sulphides (H_2S)** : In qualitative analysis of basic radical (group separation)
- 12. Sulphur (S)** : Antiseptics, vulcanization of rubber, gun powder, medicine.
- 13. Ammonia (NH_3)** : As reagent in ice factory.
- 14. Phosphorous** : (i) Red (P_4) refrigerent, in match industry etc. (ii) White (P_4) – Rat killing Medicine.

- 15. Producer gas ($\text{CO} + \text{N}_2$) :** (i) In heating furnace (ii) Cheap fuel (iii) In Extraction of metal
- 16. Water gas ($\text{CO} + \text{H}_2$) :** (i) As fuel (ii) Welding work
- 17. Coal gas :** (i) As fuel (ii) Inert atmosphere
- 18. Nitrous oxide (N_2O) :** Laughing gas, Surgery.
- 19. Carbondioxide (CO_2) :** Sodawater, Fire extinguisher.
- 20. Carbon monoxide (CO) :** In phosgene gas (COCl_2).
- 21. Graphite :** As electrodes.
- 22. Diamond :** Ornaments, Glass cutting, Rock drilling.
- 23. Alum [$\text{K}_2\text{SO}_4 \text{ Al}_2 (\text{SO}_4)_3 \cdot 24 \text{H}_2\text{O}$] :** (i) Purification of water (ii) Leather industry.
- 24. Aluminium sulphate [$\text{Al}_2(\text{SO}_4)_3 \cdot 18\text{H}_2\text{O}$] :** In paper industry/fire extinguisher.
- 25. Anhydrous aluminium chloride (AlCl_3) :** Cracking of petroleum.
- 26. Mercuric Chloride (HgCl_2) :** Calomel, Insecticides (Corrosive sublimate)
- 27. Mercuric oxide (HgO) :** Ointment, poison.
- 28. Mercury (Hg) :** Thermometer vermilion, amalgam.
- 29. Zinc Sulphide (ZnS) :** White pigment.
- 30. Zinc Sulphate (White vitriol) ($\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$) :** Lithopone, Eye ointment.
- 31. Zinc Chloride (ZnCl_2) :** Textile industry.
- 32. Zinc oxide (ZnO) :** Ointment.
- 33. Zinc (Zn) :** In battery.
- 34. Calcium carbide (CaC_2) :** Calcium cyanide & acetylene gas.
- 35. Bleaching powder [$\text{Ca}(\text{OCl}) \text{Cl}$] :** Insecticides, Bleaching actions.

- 36. Plaster of paris $[(\text{CaSO}_4)_2 \cdot 2\text{H}_2\text{O} / \text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}]$:** Statue, Surgery.
- 37. Calcium sulphate $(\text{CaSO}_4 \cdot 2\text{H}_2\text{O})$:** Cement industry.
- 38. Calcium carbonate (CaCO_3) :** Lime & toothpaste.
- 39. Copper sulphate $(\text{CuSO}_4 \cdot 5\text{H}_2\text{O})$:** Insecticides, Electric cells.
- 40. Cupric oxide (CuO) :** Blue & green glass, purification of petroleum
- 41. Cuprous Oxide (Cu_2O) :** Red glass, pesticides.
- 42. Copper (Cu) :** Electrical wire.
- 43. Sodium nitrate (NaNO_3) :** Fertilizer.
- 44. Sodium Sulphate (Glauber salt) $(\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O})$:** Medicine, cheap glass
- 45. Sodium bicarbonate (Baking soda) (NaHCO_3) :** Fire extinguisher, bakery, reagent.
- 46. Sodium Carbonate (Washing soda) :** (i) Glass industry (ii) Paper industry (iii) Removal of permanent hardness of water (iv) washing
- 47. Hydrogen peroxide (H_2O_2) :** Oxidants & reductants, Insecticides.
- 48. Heavy water (D_2O) :** Nuclear reactor.
- 49. Liquid hydrogen :** Rocket fuel.